

Sharp Weighted Four-State Hierarchy Prices for Polynomial Curvature

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Abstract

We determine the degree order of optimal two-/three-capacity hierarchy prices on four weighted source states. For nonnegative polynomial curvature of degree at most r , with arbitrary positive probabilities and posterior locations, the deterministic and stochastic worst-case prices are both $\Theta(r^2/(\log r)^2)$. The upper bound is explicitly $1936r^2/(\log r)^2$ for $r \geq 2$. Its proof selects a weighted triple using the two middle masses, compares its exact Jensen kernel with the outer-pair kernel, and combines classical Markov and Chebyshev inequalities. A full endpoint lower construction from the preceding working paper is reproduced. Separately, the classical coordinate $h = \int \sqrt{\varphi''}$ has unavoidable two-sided partition-comparison distortion $\Omega(r^2 \log r)$ for one strictly positive polynomial and one three-state source whose actual all-capacity hierarchy price is one. This limits that comparison method, not the optimal hierarchy. We also give an exact oracle equivalence for restricted indicator-ReLU splitting and show why unrestricted neuron count does not inherit it. Classical Jacobi endpoint moments identify sharp oriented Bregman asymmetry in an appendix. Arbitrary-source-count degree orders, learning guarantees, and nonvanishing absolute-risk separations remain outside the conclusions.

1 The four-state problem and the result

Separately optimal finite representations can be incompatible: an optimal three-cell partition need not refine an optimal two-cell partition. Under a strictly proper binary loss, the excess Bayes risk of a representation is a scalar Jensen distortion. Its hierarchy price is the smallest simultaneous multiplicative factor when the two-cell representation must be obtained by coarsening the three-cell one.

The preceding Phase 8 working paper [7] established a sharp $\Theta(r/\log r)$ degree envelope for four equal source masses and an $\Omega(r^2/(\log r)^2)$ lower bound when positive masses and posterior gaps may vary with degree. Its general two-capacity upper bound was $O(r^2)$. The present paper closes this logarithmic gap for exactly four states: if $\hat{C}_r^{\text{det}}$ and $\hat{C}_r^{\text{stoch}}$ denote the arbitrary-positive-weight four-state envelopes, then, for an absolute $c > 0$ and all sufficiently large r ,

$$c \frac{r^2}{(\log r)^2} \leq \hat{C}_r^{\text{stoch}} \leq \hat{C}_r^{\text{det}} \leq 1936 \frac{r^2}{(\log r)^2}. \quad (1)$$

All logarithms are natural, and r is the degree of the curvature φ'' , not of the generator φ .

The upper proof retains the original source masses in every cell kernel. An isolated weighted triple cannot support the required degree inference: even constant curvature gives a counterexample to the naive pair of three-point cost inequalities. The successful argument chooses between the two adjacent triples using their common middle masses. It then replaces the selected triple kernel by its outer-pair kernel, with error controlled by the two adjacent costs. This permits a nearby polynomial peak to force enough curvature mass in the middle-pair interval. The explicit estimate is

$$R^2 \leq 16r^2 e^{20r/\sqrt{R}} \quad (R \geq 128),$$

where R is the actual deterministic price. No lower bound on a source mass or posterior gap appears in the constants.

Domain	Established degree order or bound	Status in this paper
Four states; arbitrary positive weights; capacities two and three	$\Theta(r^2/(\log r)^2)$ in both models	Sharp order; full upper and lower proofs
Four states; all capacities	Same price as capacities two and three	One and four outputs add no cost
Arbitrary finite source count; two prescribed capacities	Prior upper $1 + 128r^2$; lower $\Omega(r^2/(\log r)^2)$ witnessed at $(2, 3)$	General degree order remains open
Arbitrary finite source count; all capacities	Prior upper $256r^2 K_r$; same lower, where $K_r = 2\log(64r^2) + 1/4$	General degree order remains open
One source and the coordinate $h = \int \sqrt{g}$	Comparison condition factor $\Omega(r^2 \log r)$	Method obstruction; example has hierarchy price one

Table 1: Distinct quantifiers. The general-source upper bounds are prior Phase 8 results, not new consequences of the four-state proof. Upper statements also bound stochastic prices; lower statements apply to both models.

Two further points prevent overinterpretation. First, a comparison of Bregman costs with squared distances may be much worse than the hierarchy itself. Section 4 realizes both sharp comparison losses for the same positive polynomial and the same three-state source. Second, an exact partition model can be embedded in a restricted indicator-ReLU class, but an unrestricted single ReLU already fits every posterior on the same one-hot domain. Section 5 therefore states an oracle equivalence with its constraints, rather than a neural-width or training theorem.

Relation to classical tools and prior work. Bregman centroids and proper-loss regret identities are classical [2, 4, 11]. Hierarchical clustering approximation has an established literature, including Lin et al. [9] and Arutyunova–Röglin [1]. The present four-state upper proof does not invoke a general hierarchy-augmentation theorem. Its polynomial ingredients are classical Markov and Chebyshev inequalities [12, 10]. Explicit squared-Chebyshev endpoint needles and $\exp(-cr\sqrt{d})$ endpoint tails occur in Slot–Laurent [13, Section 4.1], with earlier attribution there; related $O(\log^2 r/r^2)$ approximation bounds appear in [14]. The endpoint lower proof below supplies its own normalization and weighted kernel estimates and does not claim the needle construction as new.

The exact Jacobi first-moment calculation in Appendix B is a classical positive-quadrature consequence, with a direct antecedent in de Klerk–Laurent [3] and related spectral formulations in Lasserre [8]. These approximation problems do not establish partition incompatibility or the hierarchy order in (1). The focused source check is bounded, not an exhaustive originality review.

2 Distortion, channels, and hierarchy prices

Let a source index X take values in $\{1, \dots, N\}$ with probabilities $p_i > 0$, $\sum_i p_i = 1$, and posterior locations $x_i \in [0, 1]$. Unless stated otherwise, φ is differentiable and strictly convex. For polynomial results it is C^2 with nonnegative, nonzero polynomial curvature $g = \varphi''$. Such a polynomial vanishes on no nonempty interval, so φ is strictly convex even when g has isolated zeros. The oriented Bregman divergence is

$$B_\varphi(x, y) = \varphi(x) - \varphi(y) - \varphi'(y)(x - y).$$

For a partition P , set

$$D_\varphi(P) = \sum_{C \in P} D_\varphi(C), \quad D_\varphi(C) = \sum_{i \in C} p_i \varphi(x_i) - p_C \varphi(m_C), \quad p_C = \sum_{i \in C} p_i, \quad m_C = \frac{\sum_{i \in C} p_i x_i}{p_C}. \quad (2)$$

Cell costs always use original source probabilities, without conditional renormalization. Write $D(AB)$ for the cost of the cell containing states A, B , and similarly for larger cells. The flat optimum OPT_k is the minimum of $D_\varphi(P)$ over partitions with at most k cells. Refinement never increases distortion.

Lemma 2.1 (Centroids and curvature kernels). *For every cell C and report $q \in [0, 1]$,*

$$\sum_{i \in C} p_i B_\varphi(x_i, q) = D_\varphi(C) + p_C B_\varphi(m_C, q). \quad (3)$$

For $u < v$ the two orientations are

$$B_\varphi(v, u) = \int_u^v (v - t)g(t) dt, \quad B_\varphi(u, v) = \int_u^v (t - u)g(t) dt.$$

Moreover $D_\varphi(C) = \int_0^1 K_C(t)g(t) dt$, where

$$K_C(t) = \sum_{i \in C} p_i (x_i - t)_+ - p_C (m_C - t)_+ \geq 0. \quad (4)$$

Proof. Expand the Bregman expression and use $\sum_{i \in C} p_i (x_i - m_C) = 0$ to obtain (3). Twice integrating g gives the two orientation formulas and the identity $\varphi(x) = \varphi(0) + \varphi'(0)x + \int_0^1 (x - t)_+ g(t) dt$. Affine terms cancel in a Jensen gap, giving (4). Its nonnegativity is Jensen's inequality for the convex hinge $x \mapsto (x - t)_+$. $\square$

A deterministic hierarchy at capacities $k < \ell$ is a pair of partitions $H_\ell \succeq H_k$, where $\succeq$ denotes refinement and $|H_j| \leq j$. When both flat denominators are positive, its price is $\max_{j \in \{k, \ell\}} D_\varphi(H_j) / \text{OPT}_j$. The minimum over all such pairs is $\rho_{\varphi, \text{det}}$. Unless specified otherwise, four-state prices use $k = 2, \ell = 3$. There is no restriction that a hierarchical cell be contiguous in the ordered posterior locations.

For a channel $X \rightarrow Z$, define

$$D_\varphi(Z) = \mathbb{E}\varphi(x_X) - \mathbb{E}\varphi(\mathbb{E}[x_X | Z]).$$

A stochastic hierarchy is a degradation chain $X \rightarrow Z_\ell \rightarrow Z_k$, with $|Z_j| \leq j$; its optimal price $\rho_{\varphi, \text{stoch}}$ is the infimum of the same maximum ratio. The predictor is not supplied an additional public random seed.

Lemma 2.2 (The stochastic and deterministic flat optima coincide). *The infimum of $D_\varphi(Z)$ over channels with at most k outputs is OPT_k .*

Proof. Realize any channel by a random function table U , independent of X : for each input draw one output according to that channel row. Conditional on U , the map is deterministic and has at most k outputs. Conditional Jensen implies

$$\mathbb{E}\varphi(\mathbb{E}[x_X | Z]) \leq \mathbb{E}\varphi(\mathbb{E}[x_X | Z, U]).$$

Thus $D_\varphi(Z) \geq \mathbb{E}_U D_\varphi(Z | U) \geq \text{OPT}_k$. A deterministic minimizer gives the reverse inequality. $\square$

Lemma 2.3 (Two-level deterministic comparison). *For any finite source, strictly convex generator, and two output capacities with positive flat distortions,*

$$\frac{1}{2}\rho_{\varphi,\text{det}} \leq \rho_{\varphi,\text{stoch}} \leq \rho_{\varphi,\text{det}}.$$

Proof. For a stochastic degradation chain, sample independent deterministic function tables for its encoder and coarsener, independently of the source, and call the pair of tables U . Conditional on U the two representations form an admissible deterministic hierarchy. Revealing U can only reduce Jensen distortion, so at each of the two levels k ,

$$D(Z_k) \geq \mathbb{E}_U D(Z_k | U).$$

Let R be the price of this particular stochastic chain. The expected sum of the two deterministic conditional ratios is at most $2R$. At least one table therefore has sum at most $2R$, and its maximum ratio is no larger than its sum. Consequently $\rho_{\varphi,\text{det}} \leq 2R$; take the infimum over chains. Deterministic chains are themselves admissible stochastic chains, proving the other direction. No public random seed is available to the actual predictor; revealing the tables is used only in this proof. $\square$

An all-capacity deterministic hierarchy has $H_{k+1} \succeq H_k$ and $|H_k| \leq k$ for $1 \leq k \leq N$; the stochastic version is a chain $X \rightarrow Z_N \rightarrow \dots \rightarrow Z_1$. At zero optima we use the cost-inequality definition: the price is the smallest factor c for which $D_\varphi(H_k) \leq c \text{OPT}_k$ at every capacity, and likewise for channels. In particular a zero optimum requires zero hierarchical cost. On four states, any two-/three-capacity hierarchy extends by a constant output at capacity one and the identity at capacity four, so its all-capacity price is unchanged, in either model.

For an integer $r \geq 0$, define $\widehat{C}_r^{\text{det}}$ and $\widehat{C}_r^{\text{stoch}}$ as the suprema of these four-state two-/three-capacity prices over all positive probability vectors, all locations in $[0, 1]$ with $\text{OPT}_2, \text{OPT}_3 > 0$, and all generators with nonnegative, nonzero polynomial curvature of degree at most r . The generator, masses, and locations may all vary with r . Multiplying φ by a positive constant scales all distortions equally and preserves every price; adding an affine function preserves the distortions themselves.

3 The sharp degree order for four arbitrary positive weights

The four-state arbitrary-weight degree envelope has the sharp order announced in the introduction. The proof uses the original source weights in every Jensen kernel. Choosing the appropriate one of the two adjacent triples is essential: a corresponding assertion for an arbitrary weighted triple would be false.

Theorem 3.1 (Weighted four-state degree bound). *Let $g = \varphi''$ be a nonzero polynomial of degree at most r , nonnegative on $[0, 1]$. Let four source states have arbitrary positive probabilities and posterior locations in $[0, 1]$, and suppose that both flat distortions OPT_2 and OPT_3 are positive.*

Write $R = \rho_{\varphi, \det}$ for the optimal deterministic hierarchy price at output budgets two and three. If $r \geq 1$ and $R \geq 128$, then

$$R^2 \leq 16r^2 \exp\left(\frac{20r}{\sqrt{R}}\right). \quad (5)$$

In particular, if $R \geq 256$ then

$$R(\log R)^2 \leq 1936r^2. \quad (6)$$

For every integer $r \geq 2$ this gives the explicit uniform bound

$$\rho_{\varphi, \text{stoch}} \leq \rho_{\varphi, \det} \leq 1936 \frac{r^2}{(\log r)^2}. \quad (7)$$

Consequently, in the four-state degree envelopes permitting all positive weights and all posterior locations, both prices have sharp order

$$\hat{C}_r^{\det} = \Theta\left(\frac{r^2}{(\log r)^2}\right), \quad \hat{C}_r^{\text{stoch}} = \Theta\left(\frac{r^2}{(\log r)^2}\right).$$

The matching lower bounds follow from the endpoint construction, reproduced in Theorem A.1.

Proof. We first record the two elementary polynomial estimates used below. If $f \geq 0$ is a polynomial of degree at most $r \geq 1$ on an interval E of length $L > 0$, and $M_E = \max_E f$, then the rescaled Markov inequality [12] gives

$$\int_E f(t) dt \geq \frac{LM_E}{8r^2}. \quad (8)$$

Indeed, $\|f'\|_\infty \leq 2r^2 M_E/L$. From a maximizer there is an available one-sided segment of length $L/(4r^2)$, and on this segment $f \geq M_E/2$.

If t_0 lies to the right of E , at distance d from its right endpoint, the classical Chebyshev extremal inequality gives

$$|f(t_0)| \leq M_E T_r(1 + 2d/L) \leq M_E \exp\left(2r\sqrt{d/L}\right). \quad (9)$$

Here the second estimate follows from $T_r(\cosh s) = \cosh(rs)$ and $\text{arcosh}(1+x) \leq \sqrt{2x}$. To see the first estimate directly, rescale E to $[-1, 1]$ and divide by M_E . If a degree-at-most- r polynomial F bounded by one there had $|F(z)| > T_r(z)$ at $z > 1$, change its sign if necessary and set $\lambda = T_r(z)/F(z) \in (0, 1)$. At the $r+1$ alternating extrema of T_r , the polynomial $T_r - \lambda F$ has the same strict alternating signs as T_r , giving r roots in $[-1, 1]$, as well as its root at z . This contradicts its degree. These estimates are classical polynomial tools, not new approximation results.

The sole optimizer conflict is still the middle pair. Positive OPT_3 implies that the four posteriors are distinct; label them $A < B < C < D$. For a cell or pair, retain its original source weights without conditional renormalization. Put

$$p_L = D(AB), \quad q = D(BC), \quad p_R = D(CD), \quad Q = p_L + p_R, \quad T_L = D(ABC), \quad T_R = D(BCD).$$

Contiguous flat optimizers exist for arbitrary positive weights: for fixed centers $a < b$, the difference $B_\varphi(x, a) - B_\varphi(x, b)$ is affine in x with positive slope, so nearest-center assignments are contiguous, and updating to weighted centroids cannot increase distortion. Strict convexity and distinct posteriors ensure that the optimum fills its budget.

For every weighted triple, its kernel dominates the two constituent adjacent-pair kernels on their respective disjoint interiors. One way to see the domination is to merge a pair first and then merge

its weighted centroid with the remaining state; the additional Jensen gap is nonnegative, including for each hinge generator. Thus

$$T_L \geq p_L + q, \quad T_R \geq q + p_R.$$

If AB is a cheapest adjacent pair, the only incompatible contiguous coarse candidate is $A \mid BCD$; its cost obeys $T_R \geq q + p_R \geq p_L + p_R = Q$, so a compatible flat-optimal coarse partition can be chosen. The case of CD is its reflection. If BC is cheapest and either triple split is flat optimal, it too has compatible flat optima. Hence either $R = 1$, or we may work with the sole conflict

$$\text{OPT}_3 = q, \quad \text{OPT}_2 = Q,$$

where the optimal fine pair is BC and the optimal coarse split is $AB \mid CD$.

The hierarchy retaining that coarse split and its cheaper outer fine pair, and the two hierarchies retaining the fine pair BC and an adjacent triple, give the admissible-candidate inequalities

$$R \leq \frac{\min(p_L, p_R)}{q}, \quad R \leq \frac{T_L}{Q}, \quad R \leq \frac{T_R}{Q}. \quad (10)$$

They use feasible deterministic hierarchies only. They do not require an exhaustion of optimal noncontiguous hierarchies.

Choose the triple by the middle masses. Reflect the ordered configuration if necessary so that $p_B \leq p_C$. Use the triple BCD , and affinely normalize its own hull to $[0, 1]$:

$$B = 0, \quad C = c \in (0, 1), \quad D = 1, \quad (p_B, p_C, p_D) = (u, v, w), \quad 0 < u \leq v, \quad w > 0.$$

The fourth point is not used in the polynomial estimates. If the original triple hull has length L_0 , the transformed curvature is $L_0^2 g(B + L_0 t)$, preserving the curvature degree, nonnegativity, and all the costs under discussion. Denote it again by g .

Let K be the triple kernel, let k_q and k_p be its two adjacent-pair kernels, and let H be the kernel of its two outer states with their original weights:

$$\begin{aligned} K(t) &= \min\{ut + v(t - c)_+, w(1 - t) + v(c - t)_+\}, \\ k_q(t) &= \min\{ut, v(c - t)\} \mathbf{1}_{[0, c]}(t), \\ k_p(t) &= \min\{v(t - c), w(1 - t)\} \mathbf{1}_{[c, 1]}(t), \\ H(t) &= \min\{ut, w(1 - t)\}. \end{aligned}$$

In particular $T := T_R = \int_0^1 K g$, $q = \int_0^1 k_q g$, and $p_R = \int_0^1 k_p g$. Write $S = p_R + q$. By (10),

$$q \leq \frac{Q}{2R} \leq \frac{T}{2R^2}, \quad S \leq Q + q \leq \frac{T}{R} + \frac{T}{2R^2} \leq \frac{2T}{R}. \quad (11)$$

Two pointwise facts make these estimates uniform in the weights. First,

$$0 \leq K - H \leq k_q + k_p. \quad (12)$$

For $t < c$, the difference is $\min\{ut, w(1 - t) + v(c - t)\} - \min\{ut, w(1 - t)\}$, which lies between zero and $\min\{ut, v(c - t)\} = k_q(t)$. For $t > c$, interchange the two sides to obtain the bound by $k_p(t)$. Second, for $t < c$, $K(t) \leq ut$ gives

$$\frac{k_q(t)}{K(t)} \geq \min\left\{1, \frac{v(c - t)}{ut}\right\} \geq \min\left\{1, \frac{c - t}{t}\right\}.$$

For $t > c$, put $\Delta = t - c$. The elementary inequality $\min\{B, C\}/\min\{A + B, C\} \geq B/(A + B)$, for positive A, B, C , gives

$$\frac{k_p(t)}{K(t)} \geq \frac{v\Delta}{ut + v\Delta} \geq \frac{\Delta}{c + 2\Delta}.$$

Endpoint values follow by continuity; ratios need only be used inside $(0, 1)$, where $K > 0$.

A large peak and a nearby interval with small curvature cost. Assume $R \geq 128$ and put

$$\varepsilon = \frac{16}{R} \leq \frac{1}{8}, \quad I = \{t \in [0, 1] : |t - c| \leq \varepsilon c\}.$$

The preceding pointwise inequalities imply, outside I ,

$$k_q + k_p \geq \frac{\varepsilon}{1 + 2\varepsilon} K.$$

Consequently

$$\int_{[0,1] \setminus I} Kg \leq \frac{1 + 2\varepsilon}{\varepsilon} S \leq \frac{5T}{32}.$$

By (12) and (11),

$$\int_I Hg \geq T - \frac{5T}{32} - S \geq \frac{53T}{64} \geq \frac{3T}{4}.$$

On I , $H(t) \leq ut \leq u(1 + \varepsilon)c$, and $|I| \leq 2\varepsilon c$. Thus there exists $t_0 \in I$ such that

$$g(t_0) \geq \frac{T}{4\varepsilon uc^2}. \quad (13)$$

This estimate uses $H \leq ut$, rather than dividing by $H(c)$, which could be arbitrarily small as weights or gaps degenerate.

Consider the interval

$$E = [c/4, c(1 - 2\varepsilon)].$$

Its length is at least $c/2$. The point t_0 lies to its right at distance at most $3\varepsilon c$. On all of E , both terms in $k_q(t) = \min\{ut, v(c - t)\}$ are at least $2u\varepsilon c$: indeed $t \geq c/4 \geq 2\varepsilon c$, $c - t \geq 2\varepsilon c$, and $v \geq u$. With $M_E = \max_E g$, the extrapolation estimate (9) and (13) give

$$M_E \geq g(t_0) \exp(-\sqrt{24}r\sqrt{\varepsilon}) \geq \frac{T}{4\varepsilon uc^2} \exp\left(-\frac{20r}{\sqrt{R}}\right),$$

where $4\sqrt{24} < 20$. The mass estimate (8) now yields

$$\begin{aligned} q &\geq 2u\varepsilon c \int_E g(t) dt \\ &\geq 2u\varepsilon c \frac{cM_E}{16r^2} \geq \frac{T}{32r^2} \exp\left(-\frac{20r}{\sqrt{R}}\right). \end{aligned}$$

Combining this with $q \leq T/(2R^2)$ proves (5).

Extract the degree order. Set $z = r/\sqrt{R} > 0$. Equation (5) implies

$$\log(R/16) \leq 20z + 2\log z \leq 22z.$$

For $R \geq 256$, $\log(R/16) \geq \frac{1}{2} \log R$, so $\sqrt{R} \log R \leq 44r$, proving (6). For $r \geq 2$, if $R \geq 256$ and $R > r$, substitute $\log R \geq \log r$ to obtain (7). If $R \leq r$, use $(\log r)^2 \leq r$; if $R < 256$, use $(\log r)^2 \leq r^2$ and $256 < 1936$. Every deterministic hierarchy is stochastic admissible, with the same flat denominators, so the stochastic upper bound follows. The endpoint lower construction in Appendix A gives the two matching lower orders. $\square$

Quantifiers, normalization, and boundaries. The constant is independent of all positive source probabilities and all four distinct posterior locations. The proof does not impose a lower bound on a posterior gap or a source mass. All cell kernels use the original source weights; the selected triple is never conditionally renormalized. Only its posterior coordinate is rescaled, with the corresponding squared length factor in the curvature.

The matching lower witnesses can be placed in $[1/4, 3/4]$ and represented by strictly proper losses in $[0, 1/2]$, exactly as in Theorem A.1. For its notation $d = (4 \log n/n)^2$, $b = d^2$, and $M_n = \max_{[-1/2, 3/2]} g_n$, the bounded-loss affine lift multiplies every distortion by $1/(4M_n)$. The explicit feasible flat bounds therefore are

$$0 < \widetilde{\text{OPT}}_3 \leq \frac{d^4}{4M_n}, \quad 0 < \widetilde{\text{OPT}}_2 \leq \frac{2d^3}{M_n}.$$

Since $M_n \geq \int_0^1 g_n = 2 + d^2$, both upper bounds tend to zero. The sharp hierarchy-price statement is multiplicative in excess risk; it does not supply nonvanishing absolute excess costs after bounded-loss normalization. No loss normalization is needed for the upper bound.

This theorem closes the gap only for four source states and budgets two and three. On four states the all-capacity problem has the same optimum, because the one-output and four-output levels are respectively the constant representation and the identity. The proof does not improve the existing $O(r^2)$ theorem for arbitrary finite source counts and two prescribed capacities, nor the $O(r^2 \log(r+2))$ theorem for one hierarchy covering all capacities on arbitrary finite sources. It makes no claim of optimal leading constants or historical priority.

3.1 Why a triple must be selected using all four states

The preceding proof selects its triple according to the middle masses. The following constant-curvature example explains why that step cannot be discarded. Write P_{UV} for a two-point Jensen distortion and T_{UVW} for the three-point distortion, always with the original weights.

Proposition 3.2 (The two three-point inequalities alone give no bound). *Let $q = P_{BC}$, $S = P_{AB} + P_{BC}$, and $T = T_{ABC}$. There is no finite degree-only bound on a parameter R from the inequalities*

$$q \leq T/R^2, \quad S \leq 2T/R$$

for arbitrary positive triple weights, even if the curvature is the constant polynomial $g \equiv 1$ and the locations are fixed.

Proof. Take $\varphi(x) = x^2/2$, $(A, B, C) = (0, 1/2, 1)$, and original weights $(1, \varepsilon^2, \varepsilon)$ for $0 < \varepsilon \leq 1$. Common normalization of all weights is immaterial. The exact weighted variance identities give

$$P_{AB} = \frac{\varepsilon^2}{8(1 + \varepsilon^2)}, \quad q = \frac{\varepsilon^2}{8(1 + \varepsilon)}, \quad T = \frac{\varepsilon + (\varepsilon^2 + \varepsilon^3)/4}{2(1 + \varepsilon + \varepsilon^2)}.$$

In particular $q \leq \varepsilon^2/8$, $S \leq \varepsilon^2/4$, and $T \geq \varepsilon/6$. Put $R = \varepsilon^{-1/2}$. Then

$$q \leq \varepsilon^2/8 \leq \varepsilon T = T/R^2, \quad S \leq \varepsilon^2/4 \leq \varepsilon^{3/2}/3 \leq 2\sqrt{\varepsilon} T = 2T/R.$$

As $\varepsilon \downarrow 0$, $R \rightarrow \infty$ while the degree remains zero and all locations remain fixed. $\square$

Here R is only a formal parameter, not the price of a four-state hierarchy. The example obstructs the isolated three-point inference. The proof of Theorem 3.1 instead uses both triples through the common denominator Q and selects one with $u \leq v$.

4 Sharp limitations of the curvature-coordinate comparison

The coordinate $h = \int \sqrt{g}$ is classical in separable Bregman geometry [5, 6]. The next uniform estimate is the Phase 8 comparison [7], reproduced to make the obstruction self-contained. Let $g = \varphi''$ be a nonzero nonnegative polynomial on $[0, 1]$ of degree at most $r \geq 1$. Put $K_r = 2 \log(64r^2) + 1/4$ and $h(x) = \int_0^x \sqrt{g(t)} dt$.

Lemma 4.1 (Polynomial curvature-coordinate comparison). *For every $x, y \in [0, 1]$ and both orders of the arguments,*

$$\frac{1}{K_r} |h(x) - h(y)|^2 \leq B_\varphi(x, y) \leq 8r^2 |h(x) - h(y)|^2. \quad (14)$$

The constants are uniform over the polynomial coefficients and over the location and length of the interval between x and y .

Proof. The assertion is immediate for $x = y$. Fix $u < v$, and put $L = v - u$, $\mu = \int_u^v g(t) dt$, and $M = \max_{[u, v]} g$. Both μ and M are positive. The Markov mass estimates used here are

$$M \leq \frac{8r^2 \mu}{L}, \quad \int_u^v |t - a| g(t) dt \geq \frac{L\mu}{64r^2} \quad (a \in [u, v]). \quad (15)$$

For completeness, Markov's classical polynomial inequality [12], rescaled to $[u, v]$, gives $\|g'\|_\infty \leq 2r^2 M/L$. From a maximizer there is an available one-sided segment of length $L/(4r^2)$, on which $g \geq M/2$. Integration gives the first estimate. For any $a \in [u, v]$, the mass in $\{t \in [u, v] : |t - a| < L/(32r^2)\}$ is at most $LM/(16r^2) \leq \mu/2$. The complementary mass gives the second estimate.

The two divergence orientations are

$$B_\varphi(v, u) = \int_u^v (v - t) g(t) dt, \quad B_\varphi(u, v) = \int_u^v (t - u) g(t) dt.$$

Fix either of these values and denote it by B . Measure the coordinate $s \in [0, L]$ from the endpoint where its linear kernel vanishes, and denote the resulting curvature by $\tilde{g}(s)$. Explicitly, $\tilde{g}(s) = g(v - s)$ for $B_\varphi(v, u)$, and $\tilde{g}(s) = g(u + s)$ for $B_\varphi(u, v)$. Then

$$B = \int_0^L s \tilde{g}(s) ds, \quad A := h(v) - h(u) = \int_0^L \sqrt{\tilde{g}(s)} ds.$$

Taking a to be the zero-kernel endpoint in (15) gives

$$\frac{L\mu}{64r^2} \leq B \leq L\mu, \quad M \leq \frac{512r^4 B}{L^2}. \quad (16)$$

Set $\delta = L/(64r^2) \in (0, L)$ and split $A = A_0 + A_1$ at $s = \delta$. The short endpoint contribution satisfies

$$A_0^2 = \left(\int_0^\delta \sqrt{\tilde{g}(s)} ds \right)^2 \leq \delta^2 M \leq \frac{B}{8}.$$

Weighted Cauchy–Schwarz gives

$$A_1^2 = \left(\int_\delta^L \sqrt{s \tilde{g}(s)} s^{-1/2} ds \right)^2 \leq \left(\int_\delta^L s \tilde{g}(s) ds \right) \left(\int_\delta^L \frac{ds}{s} \right) \leq B \log(64r^2).$$

Therefore $A^2 \leq 2A_0^2 + 2A_1^2 \leq K_r B$, which proves the left inequality in (14).

For the other direction, $\sqrt{g(t)} \geq g(t)/\sqrt{M}$ implies

$$A^2 \geq \frac{\mu^2}{M} \geq \frac{L\mu}{8r^2}.$$

Together with $B \leq L\mu$, this proves $B \leq 8r^2 A^2$. Every estimate applies to either orientation, completing the proof. $\square$

For later use, define the transformed partition variance by

$$V_h(P) = \sum_{C \in P} \sum_{i \in C} p_i (h(x_i) - \bar{h}_C)^2, \quad \bar{h}_C = p_C^{-1} \sum_{i \in C} p_i h(x_i).$$

Minimizing the two sides of (14) over a center in each cell gives $K_r^{-1} V_h(P) \leq D_\varphi(P) \leq 8r^2 V_h(P)$. This uses that $h([0, 1])$ is an interval, and does not assume $h(m_C) = \bar{h}_C$.

4.1 Separate and simultaneous obstructions

Let $g \geq 0$ be a nonzero polynomial on $[0, 1]$, let $\varphi'' = g$, and put

$$\begin{aligned} h(x) &= \int_0^x \sqrt{g(t)} dt, & A &= h(1) - h(0), \\ B_- &= B_\varphi(0, 1) = \int_0^1 t g(t) dt, & B_+ &= B_\varphi(1, 0) = \int_0^1 (1-t) g(t) dt. \end{aligned}$$

The orientation in these definitions is intentional. Define the two degree-uniform comparison constants

$$L_r = \sup_{\substack{g \geq 0, g \neq 0 \\ \deg g \leq r}} \sup_{u \neq v} \frac{|h(u) - h(v)|^2}{B_\varphi(u, v)}, \quad U_r = \sup_{\substack{g \geq 0, g \neq 0 \\ \deg g \leq r}} \sup_{u \neq v} \frac{B_\varphi(u, v)}{|h(u) - h(v)|^2},$$

where $u, v \in [0, 1]$ and each occurrence of h and φ belongs to the same g inside its supremum.

Proposition 4.2 (The inverse logarithm cannot be removed). *For every integer $m \geq 2$, there is strictly positive polynomial curvature of degree $2m - 2$ for which*

$$\frac{A^2}{B_-} = \frac{H_m^2}{2H_m - H_{2m}} \sim \log m, \quad H_m = \sum_{j=1}^m \frac{1}{j}.$$

Consequently $L_r = \Theta(\log(r+2))$ by Lemma 4.1. In particular no absolute C can give $|h(u) - h(v)|^2 \leq C B_\varphi(u, v)$ for all polynomial degrees.

Proof. Set

$$q_m(t) = \sum_{j=0}^{m-1} (1-t)^j = \frac{1 - (1-t)^m}{t}, \quad g_m(t) = q_m(t)^2,$$

with the polynomial continuation $q_m(0) = m$. On $[0, 1]$, $q_m \geq 1$, so $\sqrt{g_m} = q_m$ and $A = H_m$. For every integer $k \geq 1$,

$$\int_0^1 \frac{1 - (1-t)^k}{t} dt = \sum_{j=0}^{k-1} \int_0^1 (1-t)^j dt = H_k.$$

The identity $[1 - (1-t)^m]^2 = 2[1 - (1-t)^m] - [1 - (1-t)^{2m}]$ therefore yields $B_- = 2H_m - H_{2m}$. Finally $H_m = \log m + O(1)$ and $H_{2m} - H_m = \log 2 + o(1)$, proving the asymptotic. Taking $m = \lfloor r/2 \rfloor + 1$ supplies a witness for every degree-at-most- r envelope, rather than only an even subsequence. $\square$

Proposition 4.3 (The forward quadratic loss cannot gain a logarithm). *For every integer $m \geq 2$, there is nonnegative nonzero polynomial curvature of degree $4m - 4$ such that*

$$\frac{B_+}{A^2} \geq \frac{m^2}{32768}.$$

Consequently $U_r = \Theta((r+1)^2)$ by Lemma 4.1. In particular an absolute bound $B_\varphi(u, v) \leq Cr^2 |h(u) - h(v)|^2 / \log(r+2)$ is false. The same asymptotic obstruction holds if curvature is required to be strictly positive.

Proof. Let T_m be the Chebyshev polynomial characterized by $T_m(\cos \theta) = \cos(m\theta)$. The quotient

$$F_m(t) = \frac{1 - T_m(1 - 2t)}{2m^2 t}, \quad F_m(0) = 1,$$

is a polynomial of degree $m - 1$. Writing $\theta = \arcsin \sqrt{t}$ shows that

$$F_m(t) = \left(\frac{\sin(m\theta)}{m \sin \theta} \right)^2, \quad 0 \leq F_m(t) \leq \min \left\{ 1, \frac{1}{m^2 t} \right\}.$$

Here $|\sin(m\theta)| \leq m \sin \theta$ follows from the geometric-sum formula for $\sin(m\theta)/\sin \theta$, and the other bound uses $|\sin(m\theta)| \leq 1$.

Set $g_m = F_m^4$, of degree $4m - 4$. Since $F_m \geq 0$,

$$A = \int_0^1 F_m(t)^2 dt \leq \frac{1}{m^2} + \int_{1/m^2}^1 \frac{dt}{m^4 t^2} = \frac{2}{m^2} - \frac{1}{m^4} \leq \frac{2}{m^2}.$$

If $0 \leq t \leq 1/(16m^2)$, then $\sqrt{t} \leq \theta \leq 2\sqrt{t}$ and $m\theta \leq 1/2$. The elementary estimate $\sin y \geq y/2$ for $0 \leq y \leq 1/2$ therefore gives $F_m(t) \geq 1/4$. On this same interval $1 - t \geq 1/2$, whence

$$B_+ \geq \int_0^{1/(16m^2)} (1 - t) F_m(t)^4 dt \geq \frac{1}{8192m^2}.$$

Division by $A^2 \leq 4/m^4$ proves the claimed bound. Taking $m = \lfloor r/4 \rfloor + 1$ covers every sufficiently large degree-at-most- r envelope.

For strict positivity one may use $\tilde{g}_m = F_m^4 + m^{-8}$, without changing the degree. Since $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$, its coordinate length is at most $2/m^2 + m^{-4} \leq 3/m^2$, whereas its B_+ is at least $1/(8192m^2)$. Thus its ratio is at least $m^2/73728$. Both examples can be rescaled to have curvature at most one; positive curvature scaling changes B and A^2 by the same factor. $\square$

Proposition 4.4 (Both losses occur for one generator and one source). *There are strictly positive polynomial curvatures of degree $r = 4m - 4$ for which the product of the best two comparison constants in the coordinate $h = \int \sqrt{g}$ is $\Omega(r^2 \log r)$. This remains true for the best two-sided partition-cost comparison constants chosen for a single three-point source. Nevertheless that source has deterministic and stochastic all-capacity hierarchy price exactly one.*

Proof. Using the two polynomials above, set

$$G_m(t) = m^4 F_m(t)^4 + q_m(1 - t)^2.$$

It is strictly positive on $[0, 1]$ and has degree $4m - 4$ for $m \geq 2$; when $m = 2$ both terms have degree four or two, and in every case the higher-degree coefficient is positive. Write $h' = \sqrt{G_m}$.

On the left half interval, the inequality $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and the preceding estimates imply

$$A_L := h(1/2) - h(0) \leq m^2 \int_0^1 F_m^2 + \int_0^{1/2} \frac{dt}{1-t} \leq 2 + \log 2 < 3.$$

The lower estimate $F_m \geq 1/4$ on $[0, 1/(16m^2)]$ gives

$$B_\varphi(1/2, 0) = \int_0^{1/2} (1/2 - t) G_m(t) dt \geq \frac{m^2}{16384}, \quad \frac{B_\varphi(1/2, 0)}{A_L^2} \geq \frac{m^2}{147456}.$$

On the right half, change variables $s = 1 - t$. Then

$$A_R := h(1) - h(1/2) \geq \int_0^{1/2} q_m(s) ds = H_m - \sum_{j=1}^m \frac{2^{-j}}{j} \geq H_m - \log 2.$$

For $t \geq 1/2$ we have $m^4 F_m(t)^4 \leq 16/m^4$, so

$$\begin{aligned} B_\varphi(1, 1/2) &= \int_{1/2}^1 (1 - t) G_m(t) dt \\ &\leq 2H_m - H_{2m} + \frac{2}{m^4} \leq H_m + \frac{2}{m^4}. \end{aligned}$$

Consequently

$$\frac{A_R^2}{B_\varphi(1, 1/2)} \geq \frac{(H_m - \log 2)^2}{H_m + 2/m^4} \sim \log m.$$

The two ratios for this same generator therefore have product $\Omega(m^2 \log m)$.

To transfer this obstruction to partition comparisons, consider the three locations $0, 1/2, 1$ with weights proportional to $1, \eta, \eta^2$, where $\eta > 0$. Let P_L merge the first two locations and let P_R merge the last two. The other location is a singleton in each partition. In general a two-point cell with locations x, y and weights $a, \eta a$ satisfies, as $\eta \downarrow 0$,

$$\frac{D_\varphi(\{x, y\})}{a\eta} \longrightarrow B_\varphi(y, x), \quad \frac{V_h(\{x, y\})}{a\eta} \longrightarrow |h(y) - h(x)|^2.$$

The first limit follows by differentiating $a\varphi(x) + a\eta\varphi(y) - a(1 + \eta)\varphi((x + \eta y)/(1 + \eta))$ at $\eta = 0$; the second follows from the exact weighted two-point variance. Overall normalization cancels from their quotient. It follows that

$$\frac{D_\varphi(P_L)}{V_h(P_L)} \longrightarrow \frac{B_\varphi(1/2, 0)}{A_L^2}, \quad \frac{V_h(P_R)}{D_\varphi(P_R)} \longrightarrow \frac{A_R^2}{B_\varphi(1, 1/2)}.$$

For each m , choose sufficiently small positive η so both ratios are at least half their limits. Thus any constants $c, C > 0$ valid for *every partition of this source*, $cV_h(P) \leq D_\varphi(P) \leq CV_h(P)$, necessarily satisfy $C/c = \Omega(m^2 \log m)$. This choice permits η to depend on m ; no assertion of a fixed minimum source mass is made.

Finally, for three states the single nontrivial capacity is two. Choose its flat optimum and extend by the singleton partition at capacity three and the one-cell partition at capacity one. This is a hierarchy with price one. Stochastic price is also one because its flat denominators coincide with the deterministic ones. $\square$

Exact scope of the obstruction. The separate orders of the comparison constants are sharp, and their product is sharp even when both constants can depend on the generator and source. Thus a transfer using a global two-sided comparison of all partition costs in this particular coordinate cannot improve its $O(r^2 \log r)$ distortion factor in general. A method that uses additional structure or a different coordinate is not excluded. These statements do not give any growing hierarchy lower bound: the last construction deliberately has hierarchy price one. Divergence or partition-comparison distortion and hierarchy incompatibility are distinct claims.

5 An exact finite indicator-growth bridge and its limits

This section provides a deliberately restricted population model in which function-preserving growth and the deterministic partition hierarchy problem coincide. It is a specialization of proper-loss regret and the Bregman centroid identity, not a general neural-training result or a priority claim.

Fix an integer $N \geq 2$, masses $p_i > 0$ with $\sum_{i=1}^N p_i = 1$, and posterior values $x_i \in [0, 1]$. The admissible inputs are the standard basis vectors $e_1, \dots, e_N \in \mathbb{R}^N$, and the population law is

$$\Pr(X = e_i) = p_i, \quad \Pr(Y = 1 \mid X = e_i) = x_i.$$

Let $\varphi \in C^1([0, 1])$ be strictly convex. For definiteness use the proper loss

$$\ell_\varphi(q, y) = -\varphi(q) - \varphi'(q)(y - q), \quad q \in [0, 1], \quad y \in \{0, 1\}.$$

Adding a constant depending only on y changes none of the excess risks. Define $B_\varphi(u, v) = \varphi(u) - \varphi(v) - \varphi'(v)(u - v)$. The conditional regret of report q at posterior x is exactly $B_\varphi(x, q)$, so strict convexity makes the loss strictly proper.

For a partition P of $\{1, \dots, N\}$, let $w_C \in \{0, 1\}^N$ be the indicator of $C \in P$, and use the restricted ReLU network

$$f_{P,q}(z) = \sum_{C \in P} q_C \text{ReLU}(w_C^\top z), \quad q_C \in [0, 1].$$

On every admissible input e_i , this equals the report q_C of its unique cell. Capacity counts the active leaf units $|P|$; there are no additional depth, split-search, sample, parameter-storage, or training-cost constraints. Hidden weights are constrained to these disjoint indicator vectors. Their ambient dimension N is not charged by the leaf count. For $C \in P$, put $p_C = \sum_{i \in C} p_i$ and $m_C = p_C^{-1} \sum_{i \in C} p_i x_i$, and define

$$D_\varphi(P) = \sum_{C \in P} \sum_{i \in C} p_i (\varphi(x_i) - \varphi(m_C)), \quad \text{OPT}_k = \min_{|P| \leq k} D_\varphi(P).$$

Write $R(P, q) = \mathbb{E} \ell_\varphi(f_{P,q}(X), Y)$ and $R^* = \mathbb{E} \ell_\varphi(x_X, Y)$, with $x_{e_i} = x_i$.

Theorem 5.1 (Exact oracle equivalence for indicator splitting). *For every finite population just specified, the following statements hold. For every partition P and every report vector q ,*

$$R(P, q) - R^* = D_\varphi(P) + \sum_{C \in P} p_C B_\varphi(m_C, q_C). \quad (17)$$

Thus the unique population-optimal reports are $q_C = m_C$, and the resulting excess risk is $D_\varphi(P)$.

Splitting a cell $C = C_0 \sqcup C_1$ into two nonempty children and copying its report into both children preserves the current predictor on every admissible input. If arbitrary such subset splits are permitted,

and exact population report minimization follows each birth, the smallest simultaneous risk factor of a one-leaf-to- N -leaf growth path is exactly

$$\rho_{\text{ind}} = \min_{\substack{H_{k+1} \\ \text{refines } H_k \\ |H_k| \leq k}} \inf\{c \geq 0 : D_\varphi(H_k) \leq c \text{OPT}_k \text{ for every } 1 \leq k \leq N\}. \quad (18)$$

Here the growth-path factor uses $R_k - R^*$ in place of $D_\varphi(H_k)$. The cost-inequality convention includes capacities with zero flat optimum. The minimum is over complete schedules selected with knowledge of the population; it does not assert a greedy or sample-based selection rule.

Proof. For each cell, expand and add the Bregman divergences:

$$\sum_{i \in C} p_i B_\varphi(x_i, q_C) = \sum_{i \in C} p_i (\varphi(x_i) - \varphi(m_C)) + p_C B_\varphi(m_C, q_C).$$

The linear term involving $x_i - m_C$ cancels. Summing proves (17); strict convexity proves uniqueness. The same identity holds conditional on any realized learned partition and reports, provided evaluation uses a fresh independent draw from the fixed population law above.

At a split, for all i we have $\mathbf{1}_C(i) = \mathbf{1}_{C_0}(i) + \mathbf{1}_{C_1}(i)$. Equal copied reports therefore preserve $f(e_i)$, and the old restricted function class is contained in the new one. In this implementation the same equality also holds on the nonnegative orthant; no claim of preservation on all of $\mathbb{R}^N$ is needed. Exact population refitting then attains $D_\varphi(P')$ for the refined partition P' .

Every binary split path is an admissible hierarchy, giving one inequality in (18). For the converse, take any admissible hierarchy (H_k) , which may have fewer than k cells. Let Q_N be the singleton partition. Induct backwards. If Q_{k+1} has $k+1$ cells and refines H_{k+1} , it also refines H_k . Since $|H_k| \leq k$, two of its cells lie inside the same H_k cell. Merge these two to form Q_k . Then Q_k has exactly k cells and refines H_k . Refinement and convexity give $D_\varphi(Q_k) \leq D_\varphi(H_k)$. Reversing this construction yields an admissible binary split path with no larger cost at any capacity. Oracle refitting realizes all its costs. This proves the reverse inequality, including zero optima. $\square$

If the reports before and after a split are population optimal, its gain is

$$D_\varphi(P) - D_\varphi(P') = \sum_{j=0}^1 p_{C_j} B_\varphi(m_{C_j}, m_C) \geq 0.$$

The gain is zero when the two child means coincide. A realized empirical refit instead retains the explicit nonnegative report-error term in (17). The theorem is an equivalence of population oracle problems, not a convergence or generalization theorem.

Proposition 5.2 (Neuron count and birth alone do not transfer the result). *On the same finite input domain and population law, removing the indicator constraint on hidden weights allows the Bayes predictor to be represented by a single ReLU unit. Also, even in the restricted indicator model, function-preserving birth alone need not achieve any finite factor relative to the flat optimum at the new capacity.*

Proof. Set $w = (x_1, \dots, x_N)$ and $f(z) = \text{ReLU}(w^\top z)$. Then $f(e_i) = x_i$ because $x_i \geq 0$, so its population excess risk is zero. This uses one hidden unit with output coefficient one. Therefore replacing leaf count by the width of this unrestricted ReLU class changes the flat oracle problem completely. In particular, high indicator-hierarchy prices cannot imply a width lower bound for this larger network class.

For the second claim take $N = 2$, equal masses, $x_1 = 1/4$, $x_2 = 3/4$, and squared loss $(y - q)^2$. The optimal one-leaf report is $1/2$ and its excess risk is $1/16$. After a singleton split that copies report $1/2$, the function and excess risk are unchanged, whereas $\text{OPT}_2 = 0$. Thus this birth, without a refit guarantee, violates every finite multiplicative cost inequality at capacity two. $\square$

Absolute normalization. For every $\lambda > 0$, replacing the loss and generator by $\lambda\ell$ and $\lambda\varphi$ scales every excess risk and every OPT_k by λ , preserving all hierarchy prices. If a loss already lies in $[0, 1/2]$, every $0 < \lambda \leq 1$ retains that same bounded-loss requirement. Given any finite instance and any $\epsilon > 0$, taking λ sufficiently small makes all its population excess risks below ϵ without changing its price. Consequently, a divergent normalized relative price alone cannot imply a nonvanishing absolute-risk separation under this upper-bound normalization. An exact fixed-scale normalization would need its own analysis.

A stochastic degradation chain is a different representation class: the deterministic indicator theorem does not identify its price with that of physical randomized neuron growth. The theorem requires the population posterior oracle and exact refits; it supplies no empirical memory, split-search, sample, or optimizer guarantee.

6 Finite computational checks

The computations check finite identities and examples; the theorems are proved analytically. They do not certify a global stochastic optimum, verify asymptotics from a finite range, or simulate neural training.

Exact rational weighted reduction. The script `check_weighted_reduction.py` enumerates all seven exact two-cell partitions, six exact three-cell partitions, and eighteen exact three-to-two hierarchies of a four-state source, including noncontiguous partitions. It tests 378 rational source/curvature instances with Python’s `Fractions.Fraction`. All flat-contiguity, optimizer-conflict, candidate-hierarchy, and selected-triple kernel checks passed; the recorded number of kernel checks is 12,993. The largest deterministic price in this finite test set was approximately 2.8153017180. The number is a property of the test set, not an estimate of the degree envelope. The JSON receipt records the exact rational value and the code hash.

Classical extremizer consistency. The companion script `check_exact_asymmetry.py` evaluates the explicit even- and odd-degree endpoint-moment extremizers of Appendix B for degrees zero through 33, using a higher-order Gauss–Legendre rule. All 34 checks passed, with largest observed relative discrepancy about 5.44×10^{-14} from the predicted shifted Jacobi zero. These are floating-point consistency checks, not interval certificates or a replacement for the quadrature proof.

Coordinate comparison. The script `check_coordinate_obstruction.py` evaluates Proposition 4.4 at $m \in \{4, 8, 16, 32, 64, 128, 256, 512\}$. It uses exactly the polynomial G_m , the locations $(0, 1/2, 1)$, and source weights proportional to $(1, \eta, \eta^2)$, with the explicit finite choice $\eta = 10^{-3}m^{-4}$. With P_L merging the first two points and P_R the last two, the reported quantity

$$\mathcal{K}_m = \frac{D_\varphi(P_L) V_h(P_R)}{V_h(P_L) D_\varphi(P_R)}$$

is a lower bound on any all-partition comparison factor C/c for that source. It is not a hierarchy price. The three-state hierarchy price is exactly one by the analytic argument in the proposition.

m	Curvature degree r	$\mathcal{K}_m$	Normalized OPT_2
4	12	4.37840	4.35994×10^{-14}
16	60	79.7071	7.11253×10^{-21}
64	252	1669.19	7.11649×10^{-28}
256	1020	33733.2	5.98140×10^{-35}
512	2044	149434	1.67346×10^{-38}

Table 2: Selected finite coordinate-comparison checks. Every row has actual all-capacity hierarchy price one. Curvature is divided by $m^4 + m^2$, a global upper bound on G_m , so it is at most one. The comparison ratios are unchanged by this normalization.

Integrals use Gauss–Legendre orders 96 and 192 separately on panels formed from the trigonometric zeros and dyadic endpoint subdivisions. The largest observed relative discrepancy between the two orders in the recorded quantities is 2.89×10^{-15} . This checks numerical stability on the finite set; it is not an interval certificate. The very small flat costs in Table 2 also show why comparison ratios should be accompanied by their absolute scales. An independent check expands the Chebyshev recurrence into polynomial coefficients at $m = 4, 16$, evaluates the Bregman moments by polynomial antiderivatives, and integrates $\sqrt{G_m}$ adaptively with 100-digit arithmetic. This independent check covers the four basic interval quantities (two coordinate lengths and two directed divergences); it does not separately recompute the tiny weighted partition costs. Its largest relative discrepancy from the panel quadrature is 2.25×10^{-15} ; it too is not interval arithmetic.

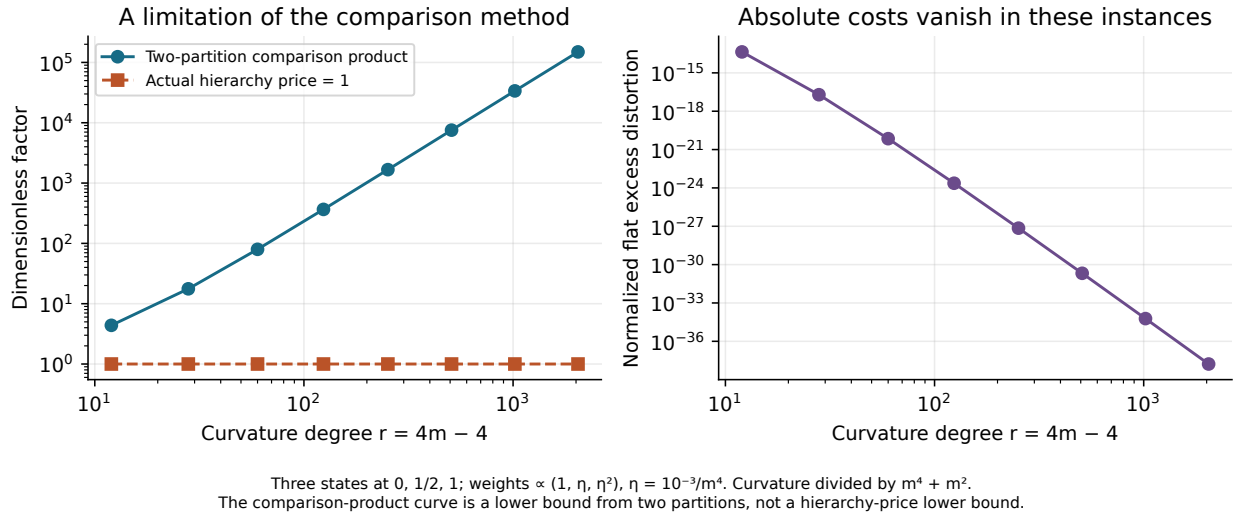


Figure 1: The same finite examples: the two-partition comparison product grows while the actual hierarchy price stays one (left), and the normalized flat distortion becomes very small (right). Lines connect computed points; no asymptotic fit is used.

Exact finite indicator-oracle checks. The script `check_indicator_bridge.py` uses exact rational arithmetic for 27 finite populations. All 108 population-oracle states, 81 birth/refit

transitions, and 27 unrestricted one-unit Bayes fits pass their identity checks. The birth-only example retains excess risk $1/16$ after copying the old report, while the new flat optimum is zero, exactly as in Proposition 5.2. These computations evaluate known population posteriors and algebraic refits; they are not training or generalization experiments.

Reproducibility and evidence boundary. The source bundle includes the scripts, JSON results, execution logs, environment details, and input hashes under `research/experiments` and `research/prior_art`. Every experiment uses finite calculations on the stated examples. The explicit lower proof and the upper proof do not depend on the computational checks. No external training run, package installation, or modification of an earlier research phase is needed to read or build this paper.

7 Remaining questions and interpretation

Theorem 3.1 and Appendix A settle the arbitrary-positive-weight four-state degree order. They rule out an $\Omega(r^2)$ construction in this exact domain, whether based on Chebyshev needles, Jacobi kernels, or another nonnegative polynomial. They do not rule out that order for more source states or different capacity pairs. Both the arbitrary-source-count two-capacity order and the arbitrary-source-count all-capacity order in Table 1 remain unresolved.

The coordinate example has a different implication. Its two-sided comparison condition factor is already $\Omega(r^2 \log r)$ after choosing the best constants for a single generator and source, while the source’s hierarchy price is exactly one. Thus improving a proof based only on that global partition comparison requires extra structure or a different coordinate. The example is no lower bound on the price of optimal hierarchies and does not prove necessity of the logarithm in the general all-capacity upper bound.

The restricted neural model is similarly exact but narrow: it identifies population oracle costs when hidden units are indicators of a partition, arbitrary subset splits are allowed, and reports are refit exactly. Removing those restrictions changes the flat oracle class. No SGD, Adam, finite-sample, continuous-domain, computational-efficiency, or generalization result is deduced. Finally, the bounded-loss lower witnesses have vanishing flat excess risks. Large multiplicative prices therefore do not establish a nonvanishing absolute-risk penalty.

Review and provenance. The upper theorem, its weighted kernel inequalities, and all displayed constants received an independent internal mathematical review. The coordinate obstruction and the two bridge/asymmetry notes also received independent internal checks. The source bundle records the exact reviewed file hashes and the arithmetic evidence. Classical approximation tools and prior Phase 8 arguments are identified explicitly; the appendix lower construction is reproduced with attribution. The focused source ledger documents access limits: original Kroó sources in the needle attribution chain were not read in this pass, and the earlier original-Lin access limitation remains. These checks establish neither external peer review nor historical priority.

A The matching endpoint lower construction

This appendix reproduces the Phase 8 endpoint construction [7], with its full kernel proof and bounded-loss normalization. Combined with Theorem 3.1, it proves the matching sharp lower order.

The source weights and posterior gaps are both allowed to vary with degree.

A.1 An endpoint-localized polynomial curvature

The only approximation ingredient needed is elementary external growth of Chebyshev polynomials. It is a classical polynomial localization mechanism; the identity $T_n(\cosh v) = \cosh(nv)$ is recorded in [10, Eq. 18.5.1]. Squared-Chebyshev endpoint needles and their exponential tail mechanisms also appear explicitly in Slot–Laurent [13, Section 4.1]; see also [14]. None of this polynomial technology is claimed as original here. The kernel estimates below are supplied in full.

For an integer n sufficiently large, put

$$d = d_n = \left(\frac{4 \log n}{n} \right)^2, \quad p_n(t) = T_n \left(\frac{2t - 1 - d}{1 - d} \right)^2, \quad Z_n = \int_0^1 p_n(t) dt, \quad \kappa_n(t) = \frac{p_n(t)}{Z_n}. \quad (19)$$

Here κ_n denotes its polynomial on the whole real line, although its normalization and probability-mass estimates refer only to $[0, 1]$. Its degree is $2n$ and it is nonnegative everywhere. For $t \in [d, 1]$ the argument of T_n belongs to $[-1, 1]$, so $p_n(t) \leq 1$. On $[0, d/2]$ the absolute value of that argument is at least $1 + d/(1 - d) \geq 1 + d$. For $0 \leq v \leq 1$, $\operatorname{arcosh}(1 + v) \geq \sqrt{v}$: for example, $\cosh(\sqrt{v}) \leq 1 + v$ follows by its power series. Using the parity of T_n and the hyperbolic identity therefore gives, for $d \leq 1/4$,

$$p_n(t) \geq \frac{1}{4} e^{2n\sqrt{d}} \quad (0 \leq t \leq d/2), \quad Z_n \geq \frac{d}{8} e^{2n\sqrt{d}}.$$

Consequently

$$\int_d^1 \kappa_n(t) dt \leq \tau_n, \quad \tau_n := 8d^{-1} e^{-2n\sqrt{d}} = \frac{1}{2n^6 (\log n)^2} = o(d^3). \quad (20)$$

The normalized maximum also satisfies

$$\max_{[0,1]} \kappa_n \leq 32n^2. \quad (21)$$

For completeness, the rescaled classical Markov inequality [12] gives $\|\kappa'_n\|_\infty \leq 2(2n)^2 M$ when $M = \|\kappa_n\|_\infty$. From a maximizer, one direction remains in $[0, 1]$ for length $1/(4(2n)^2)$, throughout which $\kappa_n \geq M/2$. Integrating and using $\int_0^1 \kappa_n = 1$ yields $M \leq 8(2n)^2$.

Define

$$g_n(t) = \kappa_n(t) + \kappa_n(1 - t) + d^2, \quad \varphi_n'' = g_n, \quad \varphi_n'(1/2) = \varphi_n(1/2) = 0. \quad (22)$$

This curvature is strictly positive on the whole real line and symmetric about $1/2$. Its degree is exactly $r = 2n$: both squared-polynomial terms have positive leading coefficient of the same even degree. The generator is a symmetric strictly convex polynomial of degree $2n + 2$.

A.2 All deterministic hierarchies are expensive

Take the sorted source locations and probabilities

$$(x_A, x_B, x_C, x_D) = (0, 2d, 1 - 2d, 1), \quad (p_A, p_B, p_C, p_D) = (a, b, b, a), \quad b = d^2, \quad a = \frac{1}{2} - b. \quad (23)$$

For all sufficiently large n , these are distinct and all weights are positive. Every distortion and flat denominator below uses these original weights, with no conditional renormalization of a cell.

For a pair $u < v$ of respective weights p_u, p_v , its exact Jensen curvature kernel is

$$K_{uv}(t) = \min\{p_u(t - u), p_v(v - t)\} \mathbf{1}_{[u,v]}(t). \quad (24)$$

For an arbitrary cell S , the kernel is

$$K_S(t) = \sum_{i \in S} p_i(x_i - t)_+ - p_S(m_S - t)_+, \quad D(S) = \int K_S(t) g_n(t) dt.$$

These formulas follow by writing a twice differentiable convex generator, up to an affine term, as an integral of $(x - t)_+$ against its curvature.

Set

$$P = D(AB) = D(CD), \quad q = D(BC), \quad T = D(ABC) = D(BCD), \quad J = D(AD).$$

The equalities use reflection symmetry. Since $128n^2b = 128n^2d^2 \rightarrow 0$, equations (20)–(21) imply, eventually,

$$I = [4b, d] \neq \emptyset, \quad \int_I \kappa_n(t) dt \geq 1 - \tau_n - 128n^2b \geq \frac{1}{2}. \quad (25)$$

We may also assume $d \leq 1/16$ and $\tau_n \leq d^2/2$.

The pair AB has mean $4bd$. Thus on I , which lies above that mean, $K_{AB}(t) = b(2d - t) \geq bd$. Conversely $K_{AB} \leq 2bd$ everywhere. On $[0, 2d]$, the reflected needle contributes at most τ_n to the integrated curvature, since $1 - 2d \geq d$. Hence

$$\int_0^{2d} g_n \leq 1 + \tau_n + 2d^3 \leq 2.$$

Together with (25), this proves

$$\frac{bd}{2} \leq P \leq 4bd. \quad (26)$$

For the middle pair, $K_{BC} \leq b/2$. The two needles contribute at most $2\tau_n$ on $[2d, 1 - 2d]$, and the constant curvature contributes at most d^2 . Therefore

$$0 < q \leq \frac{b}{2}(2\tau_n + d^2) \leq bd^2. \quad (27)$$

The triple ABC has mean $b/(1/2 + b) \leq 2b$. For $t \in I$ this mean lies below t , while both inner locations lie above t , so

$$K_{ABC}(t) = b(2d - t) + b(1 - 2d - t) = b(1 - 2t) \geq b/2.$$

The outer pair has kernel $K_{AD}(t) = a \min(t, 1 - t)$. On I this is at least $4ab \geq b$, because $a \geq 1/4$. Thus

$$T \geq b/4, \quad J \geq b/2. \quad (28)$$

We now cover all deterministic hierarchy choices, including noncontiguous ones. A hierarchy with fewer than the allowed numbers of cells can be refined to exactly three fine cells and two coarse cells without increasing either cost: split fine cells within their current coarse cells, then split the coarse level if necessary by grouping the three fine cells. Every exact three-cell partition of four states merges one pair. Besides BC , each pair has cost at least P . This is immediate for AB, CD ; the pair AC has the same endpoint weights as AB and a more distant right endpoint, so its kernel dominates K_{AB} pointwise, and BD is its reflection. Finally, $J \geq b/2 \geq P$ by (26) and $d \leq 1/16$.

If the fine pair is not BC , then $\text{OPT}_3 \leq q$ gives fine-level ratio at least $P/q \geq 1/(2d)$. If the fine pair is BC , its exact two-cell coarsenings are $ABC \mid D$, $A \mid BCD$, and $AD \mid BC$. By (28) each costs at least $b/4$. Since the available coarse split $AB \mid CD$ gives $\text{OPT}_2 \leq 2P \leq 8bd$, this hierarchy has coarse-level ratio at least $1/(32d)$. These two cases prove

$$\rho_{\varphi_n, \text{det}} \geq \frac{1}{32d_n}, \quad \rho_{\varphi_n, \text{stoch}} \geq \frac{1}{64d_n}, \quad (29)$$

the second inequality following from Lemma 2.3. Notice that this proof needs neither exact flat optimizer identification nor an assertion that contiguous deterministic hierarchies exhaust the optimum.

Theorem A.1 (Matching arbitrary-weight four-state lower bound). *Let $\hat{C}_r^{\text{det}}$ and $\hat{C}_r^{\text{stoch}}$ denote the two/three-output four-state degree envelopes in which all positive source weights are allowed. There is an absolute $c > 0$ such that, for all sufficiently large r ,*

$$\hat{C}_r^{\text{det}} \geq \hat{C}_r^{\text{stoch}} \geq c \frac{r^2}{(\log r)^2}.$$

Consequently no absolute-constant $O(r/\log r)$ bound holds in this weighted four-state setting. The witnesses can have all posteriors in $[1/4, 3/4]$ and polynomial generators inducing smooth strictly proper losses bounded in $[0, 1/2]$.

Proof. Equations (19) and (29) give the claim at degrees $r = 2n$, since $1/d_n = n^2/(16(\log n)^2)$. For a degree-at-most- r envelope take $n = \lfloor r/2 \rfloor$.

To put all posteriors in the stated interior interval, use $y = 1/4 + x/2$. The four locations become $(1/4, 1/4 + d, 3/4 - d, 3/4)$. Let $M_n = \max_{t \in [-1/2, 3/2]} g_n(t)$ and set

$$\tilde{\varphi}_n(y) = \frac{\varphi_n(2y - 1/2)}{4M_n}.$$

The fact that g_n is a sum of squares plus d^2 on the *whole real line* ensures $0 < \tilde{\varphi}_n'' \leq 1$ on $[0, 1]$. This affine coordinate change and positive rescaling preserve every price and curvature degree. The generator remains symmetric about $1/2$. Put $h(y) = \tilde{\varphi}_n(0) - \tilde{\varphi}_n(y)$ and define

$$\ell(y, 1) = h(y) + (1 - y)h'(y), \quad \ell(y, 0) = h(y) - yh'(y).$$

Expected regret is $B_{\tilde{\varphi}_n}$, so the loss is strictly proper. Symmetry and the supporting-line inequalities make both losses nonnegative; their derivatives are $-(1 - y)\tilde{\varphi}_n''(y)$ and $y\tilde{\varphi}_n''(y)$, respectively. Their common maximum is $\tilde{\varphi}_n'(1) = -\tilde{\varphi}_n'(0) = \int_{1/2}^1 \tilde{\varphi}_n'' \leq 1/2$. $\square$

The smallest mass in this family is $d_n^2 = 256(\log n)^4/n^4$. The generators, weights, and posterior gaps all vary with n . This does not determine a sharp joint minimum-mass/degree envelope. The normalization preserves price, not absolute excess cost; explicit vanishing flat-cost bounds appear after Theorem 3.1.

B Exact endpoint moments and polynomial Bregman asymmetry

The endpoint first-moment problem can be solved exactly by positive quadrature. Its sum-of-squares version is already the linear-objective case in de Klerk–Laurent, *Worst-case examples for*

Lasserre's measure-based hierarchy for polynomial optimization on the hypercube, [3, Theorem 3.1 and Corollary 3.2]. The reduction below is a classical quadrature consequence, not a new approximation principle. It strengthens the constants and identifies extremizers for one ingredient; it does not determine a hierarchy price.

For an integer $r \geq 0$, define

$$\lambda_r = \inf_{\substack{g \neq 0, \deg g \leq r \\ g \geq 0 \text{ on } [0,1]}} \frac{\int_0^1 tg(t) dt}{\int_0^1 g(t) dt}.$$

For $\alpha, \beta > -1$, write $\xi_j^{(\alpha, \beta)}$ for the smallest zero of the degree- j Jacobi polynomial $P_j^{(\alpha, \beta)}$ on $[-1, 1]$, with weight $(1-x)^\alpha(1+x)^\beta$.

Theorem B.1 (Exact endpoint moment). *For every $m \geq 0$,*

$$\lambda_{2m} = \frac{1 + \xi_{m+1}^{(0,0)}}{2}, \quad \lambda_{2m+1} = \frac{1 + \xi_{m+1}^{(1,0)}}{2}. \quad (30)$$

The infima are attained. In particular, $\lambda_r = \Theta((r+1)^{-2})$.

Proof. For even degree, take the $m+1$ -node Gauss–Legendre rule on $[0, 1]$ [10, Section 3.5(v)]. Its nodes $t_1 < \dots < t_{m+1}$ are the shifted Legendre zeros, its weights ω_j are positive, and it integrates polynomials of degree at most $2m+1$ exactly. Thus for every admissible g ,

$$\int_0^1 tg(t) dt - t_1 \int_0^1 g(t) dt = \sum_{j=1}^{m+1} \omega_j (t_j - t_1) g(t_j) \geq 0.$$

The polynomial $g(t) = \prod_{j=2}^{m+1} (t - t_j)^2$ has degree $2m$, is nonnegative, and makes the right side zero. The empty product for $m=0$ is one. This proves the first formula.

For clarity, the requisite odd-degree quadrature rule is constructed here. Let J be the degree- $m+1$ polynomial orthogonal for weight $1-t$ on $[0, 1]$. Its $m+1$ simple interior zeros are $t_1 < \dots < t_{m+1}$; add the node $t_{m+2} = 1$. Integrate the Lagrange interpolant on these $m+2$ nodes to define an interpolatory rule. For a polynomial f of degree at most $2m+2$, divide

$$f(t) = (t-1)J(t)q(t) + s(t), \quad \deg q \leq m, \quad \deg s \leq m+1.$$

The first summand integrates to zero by orthogonality and vanishes at all nodes. The rule therefore integrates f exactly. If L_j is the degree- $m+1$ cardinal Lagrange polynomial, exactness for L_j^2 gives $\omega_j = \int_0^1 L_j(t)^2 dt > 0$. Applying this positive rule to g and tg proves that the moment ratio is at least t_1 . Equality is attained by

$$g(t) = (1-t) \prod_{j=2}^{m+1} (t - t_j)^2,$$

of degree $2m+1$. The orthogonality weight identifies J with the shifted $P_{m+1}^{(1,0)}$, proving the second formula.

For both fixed parameter pairs, the smallest Jacobi zero is $-1 + \Theta((m+1)^{-2})$. This standard extreme-zero estimate is recorded in de Klerk–Laurent [3, Corollary 2.3], and gives the stated order. $\square$

Theorem B.2 (Sharp oriented polynomial asymmetry). *Let $\varphi'' = g$ be a nonnegative nonzero polynomial of degree at most r on an interval $[u, v]$, $u < v$. Then*

$$\max \left\{ \frac{B_\varphi(v, u)}{B_\varphi(u, v)}, \frac{B_\varphi(u, v)}{B_\varphi(v, u)} \right\} \leq \frac{1 - \lambda_r}{\lambda_r}. \quad (31)$$

The bound is attained within nonnegative polynomial curvature, is the supremum if curvature is required to be strictly positive everywhere on the interval, and is $\Theta((r + 1)^2)$.

Proof. Affine rescaling reduces to $[0, 1]$. Writing $\mu = \int_0^1 g$ and $a = \mu^{-1} \int_0^1 tg$ gives

$$B_\varphi(0, 1) = a\mu, \quad B_\varphi(1, 0) = (1 - a)\mu.$$

The preceding theorem and its reflected version imply $\lambda_r \leq a \leq 1 - \lambda_r$. The ratio bound follows, and the extremizing curvatures above attain equality. Such curvature still induces strict convexity because it vanishes on no nonempty interval. For everywhere-positive curvature, add $\varepsilon > 0$ and send it to zero; all displayed integrals converge and the denominator stays positive. The extreme-zero order proves the last assertion. $\square$

Why oriented asymmetry is not a hierarchy lower bound. After normalizing $\int_0^1 g = 1$, a first-moment estimate $\int_0^1 tg = O(r^{-2})$ gives only $\int_d^1 g \leq O((r^2d)^{-1})$. At $d \asymp r^{-2}$ this is a constant bound, whereas the existing four-state endpoint proof uses a leakage bound much smaller than a power of d . The exact moment theorem does not supply that stronger assertion. Conversely, sharp oriented asymmetry need not generate incompatible optimal partitions at two capacities. It cannot be substituted for a hierarchy-price lower bound. The general-source $O(r^2)$ two-capacity bound in Phase 8 uses an anchor quasi-triangle inequality, a different optimization from the endpoint orientation ratio solved here. The sharp four-state result in Theorem 3.1 instead uses a direct kernel argument.

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